Why Numerical Integration is Used:
Numerical integration is used to approximate the value of definite integrals when:

1. **No analytical (exact) solution exists:** Some functions are too complex or don't have a known antiderivative.
2. **Data is discrete:** You only have data points, not a continuous function.
3. **Functions are difficult to integrate analytically:** Even if an antiderivative exists, it might be too complicated or time-consuming to compute manually.
4. **Real-time or fast computation is required:** For simulations or models where speed is more important than precision.

Common methods:

- Trapezoidal Rule
- Simpson's Rule
- Monte Carlo Integration
- Riemann Sums

Applications in Computer Science:

1. Graphics and Game Development

- Shading, lighting, and rendering (e.g., calculating the total light hitting a surface).
- Ray tracing simulations often involve Monte Carlo integration.

2. Machine Learning and AI

- Probabilistic models: Compute expectations and marginal probabilities.
- Bayesian inference: Integrals over posterior distributions.

3. Signal Processing

- Area under signal curves, filtering, and Fourier transforms.
- Integrating sampled signals to compute power or energy.

4. Computer Vision

- Calculating object shapes or surface areas from scanned point data.

5. Scientific Computing & Simulations

- Physics simulations: Motion, heat, fluid dynamics, etc.
- Engineering software (like finite element methods).

6. Robotics

- Path planning: Calculating distances or energy usage over irregular terrain.
- Sensor fusion: Estimating positions from continuous sensor readings.

7. Data Science

- Estimating the area under the curve (AUC) for performance metrics like ROC curves.

Example:
If you're simulating a robot arm's movement and want to compute the **total energy used** over time, you might not have a simple formula. But with discrete force and time data, **numerical integration** gives you a fast and approximate result:

$$\text{Energy} = \int_{t_1}^{t_2} F(t) \cdot v(t)\, dt$$